04f — HATEOAS

Key Concepts

Term	Definition
HATEOAS	Hypermedia As The Engine Of Application State — REST constraint where responses include links for available actions
Hypermedia	Links embedded in API responses that describe what the client can do next
Richardson Maturity Model	4-level model measuring how "RESTful" an API is (0–3)
Level 3	HATEOAS — highest REST maturity level

How It Works

1. Client makes a request (e.g. GET `/orders/456`)
2. Server returns data **plus** `_links` — URLs for available next actions
3. Client follows links rather than constructing URLs itself
4. Available links change based on current state (e.g. cancelled order has no `pay` link)

```
{
  "orderId": "456",
  "status": "pending",
  "_links": {
    "self":   { "href": "/orders/456" },
    "cancel": { "href": "/orders/456/cancel", "method": "DELETE" },
    "pay":    { "href": "/orders/456/pay",    "method": "POST" }
  }
}
```

Richardson Maturity Model

Level	What It Means	Example
0	One endpoint, RPC-style	`POST /api` for everything
1	Resources	`/users`, `/orders`
2	HTTP verbs + resources	`GET /users`, `DELETE /orders/1`
3	HATEOAS — hypermedia links	Response includes `_links`

Most real-world APIs are **Level 2**.

Benefits vs Reality

	Theory	Reality
Client decoupling	URL changes don't break clients — they follow links	Clients hardcode URLs anyway
Discoverability	Clients explore the API dynamically	Clients read docs, not links
State awareness	Only valid actions appear as links	Extra server-side logic to generate

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Why It's Rarely Used

- Clients hardcode URLs — the decoupling benefit is rarely realized
- Generating accurate, state-aware links server-side adds complexity
- Most teams get enough value from Level 2 (resources + HTTP verbs)
- No major public API fully implements it (most stop at Level 2)

Interview Questions

Q: What is HATEOAS? A: A REST constraint where API responses include hypermedia links describing available next actions, so clients discover the API dynamically rather than hardcoding URLs.

Q: What level is HATEOAS on the Richardson Maturity Model? A: Level 3 — the highest level of REST maturity.

Q: Why don't most APIs implement HATEOAS? A: Clients hardcode URLs anyway so the decoupling benefit is rarely realized, and generating accurate state-aware links server-side adds complexity with little practical payoff. Most APIs stop at Level 2.

Q: What is the benefit of HATEOAS in theory? A: If server URLs change, clients don't break — they just follow whatever links the response provides, decoupling client from server structure.